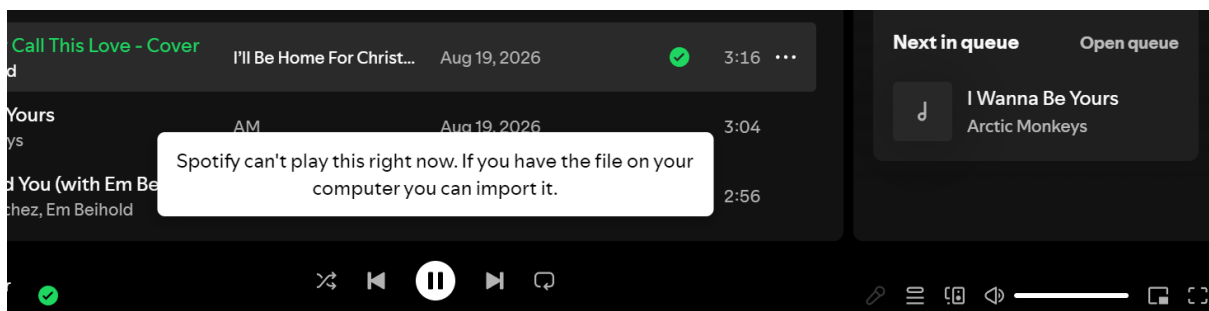
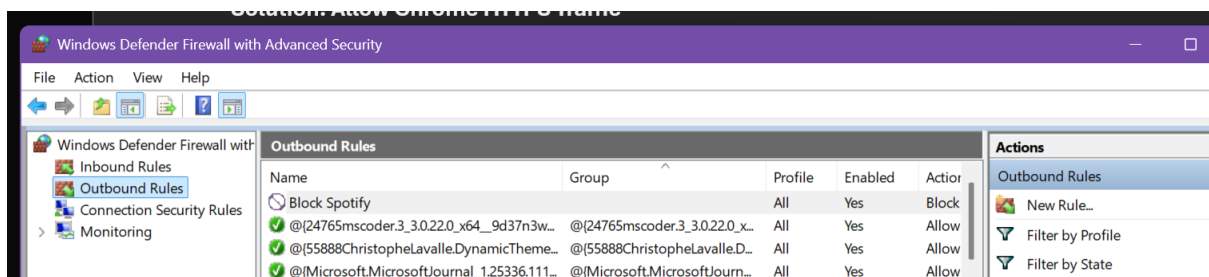


Firewall Basics

Task – 1 : Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

Solution : Block Spotify Using Windows Defender Firewall

1. Open **Windows Defender Firewall with Advanced Security**.
2. Click **Outbound Rules** → **New Rule**.
3. Select **Program** → **Next**.
4. Select **This program path** and browse to the Spotify .exe file.
5. Select **Block the connection**.
6. Keep **Domain, Private, and Public** selected.
7. Give the rule a name such as **Block Spotify**.
8. Click **Finish**.



Task – 2 : Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

Solution: Allow Chrome HTTPS Traffic

1. Open **Windows Defender Firewall with Advanced Security**.
2. Go to **Inbound Rules** → **New Rule**.
3. Select **Port** → **TCP** → enter port **443**.
4. Select **Allow the connection**.
5. Select the required network profiles.
6. Give the rule a name such as **Allow Chrome HTTPS 443**.
7. Click **Finish**.

Task – 3 : Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall. Hint : Use the 'Scope' tab when creating a new rule to specify the IP address.

Solution:

1. Open **Windows Defender Firewall with Advanced Security**.
2. Choose **Inbound Rules** or **Outbound Rules**, depending on the traffic direction.
3. Click **New Rule** → **Custom**.
4. Select **All programs**.
5. Open the **Scope** tab.
6. Under **Remote IP address**, select **These IP addresses**.
7. Click **Add** and enter:
123.45.67.89
8. Select **Block the connection**.
9. Apply it to **Domain, Private, and Public** profiles.
10. Name the rule **Block 123.45.67.89** and click **Finish**.

The image shows two screenshots from Windows Defender Firewall with Advanced Security.

The top screenshot is the 'New Inbound Rule Wizard' window, specifically the 'Scope' tab. It asks 'Which local IP addresses does this rule apply to?' with options 'Any IP address' and 'These IP addresses:'. The 'These IP addresses' option is selected, and a text box contains '123.45.67.89'. There are 'Add...', 'Edit...', and 'Remove' buttons. Below this, it asks 'Which remote IP addresses does this rule apply to?' with options 'Any IP address' and 'These IP addresses:'. The 'Any IP address' option is selected. There is a 'Customize...' button.

The bottom screenshot shows the 'Inbound Rules' list in the Windows Defender Firewall with Advanced Security console. The list includes the rule 'block 123.45.67.89' which is enabled and set to 'Block'. Other rules include 'Allow ICMP Ping', 'Allow ICMPv4', 'Allow ICMPv4 Ping', 'Allow Ping', and 'Allow Ping'.

Name	Group	Profile	Enabled	Action
block 123.45.67.89		All	Yes	Block
Allow ICMP Ping		All	Yes	Allow
Allow ICMPv4		All	Yes	Allow
Allow ICMPv4 Ping		All	Yes	Allow
Allow Ping		All	Yes	Allow
Allow Ping		All	Yes	Allow

Task – 4 : Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Solution : Hardware Firewall vs Software Firewall

Feature	Hardware Firewall	Software Firewall
Location	Separate network device	Installed on a computer
Protection	Protects multiple devices/networks	Mainly protects one device
Management	Usually managed centrally	Managed on individual systems
Use Case	Large offices and organizations	Personal computers and laptops

Example:

- **Hardware Firewall:** Used at a company's network gateway to control traffic entering and leaving the office network.
- **Software Firewall:** Windows Defender Firewall protects an individual Windows computer from unwanted network connections.